Article

The sharp universal spectral constant for scaled q-numerical ranges

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Abstract

Let A be an $n \times n$ complex matrix, $n \geq 2$, and let $0 < |q| \leq 1$. The q-numerical range and its scaled form are

$$W_q(A) = \{y^*Ax : \|x\|_2 = \|y\|_2 = 1,\ y^*x = q\}, \qquad \Omega_q(A) = q^{-1}W_q(A).$$

We prove that the sharp universal spectral constant for the family of scaled q-numerical ranges is

$$C_q = \max\left\{1, \frac{2|q|}{1+\sqrt{1-|q|^2}}\right\}.$$

Thus, for every rational function f pole-free on $\Omega_q(A)$, $\|f(A)\|$ is at most C_q times the maximum modulus of f on $\Omega_q(A)$, and C_q is the least constant valid uniformly in n and A. The proof is a transfer from the sharp numerical-range theorem. Its geometric input shows that the numerical range of every adapted rank-one stretch of A lies in the scaled q-range. Applying a numerical-range spectral-set estimate to these stretches produces a simultaneous family of Schur estimates. A disk automorphism adapted to a maximizing singular-vector pair then extracts the norm at the identity stretch. More generally, a universal numerical-range spectral constant K transfers to the constant $\max\{1, K/\kappa(q)\}$ on the scaled q-range, where $\kappa(q) = (1+\sqrt{1-|q|^2})/|q|$. The two lower-bound branches defining C_q are attained, respectively, by the constant rational function and by the identity polynomial on a 2×2 square-zero matrix.

Keywords: q-numerical range; spectral set; Crouzeix conjecture; numerical range; rank-one similarity; Schur function

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1. Introduction

For $A \in \mathbb{M}_n(\mathbb{C})$, its numerical range is

$$W(A) = \{x^*Ax : x \in \mathbb{C}^n,\ \|x\|_2 = 1\}.$$

It is compact and convex, and it contains the spectrum of A. For $C \geq 1$, a compact set $E \subset \mathbb{C}$ containing $\sigma(A)$ is a C-spectral set for A if

$$\|g(A)\| \leq C \max_{z \in E} |g(z)| \tag{1}$$

for every rational function g pole-free on E. Crouzeix conjectured that $W(A)$ is always a 2-spectral set [1], and the sharp constant 2 was established in [2].

For $n \geq 2$ and $0 < |q| \leq 1$, the q-numerical range is

$$W_q(A) = \{y^*Ax : \|x\|_2 = \|y\|_2 = 1,\ y^*x = q\}, \qquad \Omega_q(A) = \frac{1}{q}W_q(A). \tag{2}$$

The set $W_q(A)$ is compact, and Tsing proved its convexity [3]. The scaled set $\Omega_q(A)$ contains $W(A)$ and hence $\sigma(A)$. O'Loughlin and Rani formulated the polynomial inequality below as Conjecture 4.2 [4]. The theorem proves that conjecture together with its rational spectral-set extension.

Theorem 1.1 (Main theorem). *Let $n \geq 2$, let $A \in \mathbb{M}_n(\mathbb{C})$, let $0 < |q| \leq 1$, and let f be a rational function pole-free on $\Omega_q(A)$. Then*

$$\|f(A)\| \leq C_q \max_{z \in \Omega_q(A)} |f(z)|, \qquad C_q = \max\left\{1, \frac{2|q|}{1+\sqrt{1-|q|^2}}\right\}. \tag{3}$$

Equivalently, $\Omega_q(A)$ is a C_q-spectral set for A. In particular, the same inequality holds for every polynomial f.

The proof has three steps. First, Section 2 proves nesting of the scaled ranges and shows that every adapted rank-one stretch has numerical range inside $\Omega_q(A)$. Second, Section 3 proves that uniform Schur estimates over precisely these stretches force an improved norm estimate at the identity stretch. Third, Section 4 packages the two facts into a transfer principle for any universal numerical-range spectral constant K. Taking $K = 2$ gives (3); a 2×2 nilpotent matrix proves sharpness.

2. Rank-one stretch geometry

We use the inner product

$$\langle x, y \rangle = y^* x,$$

which is linear in its first variable. Thus $W_q(A) = \{\langle Ax, y \rangle : \|x\|_2 = \|y\|_2 = 1,\ \langle x, y \rangle = q\}$. Changing the phase of y in (2) gives

$$\Omega_q(A) = \Omega_{|q|}(A). \tag{4}$$

We may therefore put

$$r = |q|, \qquad \tau = \sqrt{1-r^2}, \qquad \kappa = \frac{1+\tau}{r}. \tag{5}$$

The equivalent identities

$$r = \frac{2\kappa}{\kappa^2+1}, \qquad \frac{\tau}{r} = \frac{\kappa^2-1}{2\kappa}, \qquad \frac{2}{\kappa} = \frac{2r}{1+\sqrt{1-r^2}} \tag{6}$$

will be used repeatedly.

After division by r, Tsing's Lemma 5 gives the following disk formula [3]. It will also be used in the sharpness argument. For a unit vector x, put

$$\rho_A(x) = \left(\|Ax\|_2^2 - |\langle Ax, x \rangle|^2\right)^{1/2}.$$

The radicand is nonnegative by Cauchy–Schwarz.

Lemma 2.1 (Disk formula and nesting). *Let $A \in \mathbb{M}_n(\mathbb{C})$, $n \geq 2$, and $0 < r \leq 1$. Then*

$$\Omega_r(A) = \bigcup_{\|x\|_2 = 1} \left\{ z \in \mathbb{C} : |z - \langle Ax, x \rangle| \leq \frac{\sqrt{1-r^2}}{r} \rho_A(x) \right\}. \tag{7}$$

Consequently, if $0 < r \leq s \leq 1$, then

$$\Omega_s(A) \subseteq \Omega_r(A).$$

In particular, $W(A) = \Omega_1(A) \subseteq \Omega_r(A)$.

Proof. Let x, y be unit vectors with $\langle x, y\rangle = r$, and put $e_x = Ax - \langle Ax, x\rangle x$. Then $e_x \perp x$ and $\|e_x\|_2 = \rho_A(x)$. Since $y - rx \perp x$ and $\|y - rx\|_2 = \sqrt{1-r^2}$, Cauchy–Schwarz gives

$$\left|\frac{1}{r}\langle Ax, y\rangle - \langle Ax, x\rangle\right| = \frac{1}{r}|\langle e_x, y - rx\rangle| \le \frac{\sqrt{1-r^2}}{r}\rho_A(x).$$

This proves the inclusion from left to right in (7).

Conversely, suppose that ζ belongs to the disk indexed by a unit vector x_0. On the unit sphere define

$$g(x) = |\zeta - \langle Ax, x\rangle| - \frac{\sqrt{1-r^2}}{r}\rho_A(x).$$

The function g is continuous and $g(x_0) \le 0$. A complex matrix has a unit eigenvector e, and $\rho_A(e) = 0$, so $g(e) \ge 0$. The unit sphere in $\mathbb{C}^n$ is path connected for $n \ge 2$; hence there is a unit vector u with $g(u) = 0$.

Write $\alpha = \langle Au, u\rangle$ and $d = \rho_A(u)$. Choose a unit vector $w \perp u$ such that $\langle Au, w\rangle = d$: when $d > 0$, take $w = (Au - \alpha u)/d$, and when $d = 0$, take any unit vector orthogonal to u. If $d > 0$, set

$$\beta = \frac{r\,\overline{\zeta - \alpha}}{d};$$

if $d = 0$, set $\beta = \sqrt{1-r^2}$. The equality $g(u) = 0$ gives $|\beta| = \sqrt{1-r^2}$. Thus $y = ru + \beta w$ is a unit vector with $\langle u, y\rangle = r$, and if $d > 0$, the definition of β gives $\overline{\beta} d/r = \zeta - \alpha$. If $d = 0$, then $g(u) = 0$ gives $\zeta = \alpha$. In either case,

$$\frac{1}{r}\langle Au, y\rangle = \alpha + \frac{\overline{\beta} d}{r} = \zeta.$$

This proves the reverse inclusion.

Finally, the coefficient $\sqrt{1-r^2}/r$ decreases with r on $(0, 1]$. Comparing the disks in (7) proves nesting, and the case $r = 1$ gives $\Omega_1(A) = W(A)$. $\square$

For a unit vector v, let P_v be the orthogonal projection onto $\mathbb{C}v$ and define the rank-one stretch

$$S_v = I + (\kappa - 1)P_v. \tag{8}$$

Lemma 2.2 (Rank-one stretch geometry). *Let $A \in \mathbb{M}_n(\mathbb{C})$, $n \ge 2$, and $0 < r \le 1$, with κ as in (5). For every unit vector v, $S_v \succ 0$,*

$$S_v^{-1} = I + (\kappa^{-1} - 1)P_v,$$

and

$$W(S_v A S_v^{-1}) \subseteq \Omega_r(A). \tag{9}$$

Proof. The formulas for S_v and S_v^{-1} follow from $P_v^2 = P_v$; positivity follows from $\kappa \ge 1$. Fix a unit vector u and put

$$\ell = \|S_v^{-1}u\|_2, \qquad m = \|S_v u\|_2, \qquad \theta = \|P_v u\|_2^2.$$

Orthogonality of $P_v u$ and $(I - P_v)u$ gives

$$m^2 = \kappa^2\theta + 1 - \theta, \qquad \ell^2 = \kappa^{-2}\theta + 1 - \theta.$$

Since $0 \le \theta \le 1$,

$$\ell^2 m^2 = 1 + (\kappa - \kappa^{-1})^2\theta(1-\theta) \le \frac{1}{4}(\kappa + \kappa^{-1})^2.$$

Moreover, $1 = |\langle S_v^{-1}u, S_v u\rangle| \le \ell m$. Because S_v is invertible, $\ell, m > 0$. Therefore, with $s = (\ell m)^{-1}$,

$$r \le s \le 1, \qquad \frac{1}{2}(\kappa + \kappa^{-1}) = \frac{1}{r}.$$

The vectors
$$x = \frac{S_v^{-1}u}{\ell}, \qquad y = \frac{S_v u}{m}$$
are unit vectors with $\langle x, y\rangle = s$, and
$$\langle S_v A S_v^{-1} u, u\rangle = \frac{1}{s}\langle Ax, y\rangle \in \Omega_s(A) \subseteq \Omega_r(A)$$
by Lemma 2.1. This proves (9). $\square$

3. Rank-one stretch extraction

A convenient family of disk automorphisms is
$$\varphi_{\omega,a}(z) = \frac{\omega z - a}{1 - a\omega z}, \qquad |\omega| = 1, \quad 0 \le a < 1.$$
The next lemma is independent of numerical ranges. It is the step that converts simultaneous estimates over the rank-one stretches (8) into a sharper norm bound at the identity stretch.

Lemma 3.1 (Rank-one stretch extraction). *Let $n \ge 2$, let $\kappa \ge 1$ and $K \in \mathbb{R}$, and let $X \in \mathbb{M}_n(\mathbb{C})$ satisfy $\sigma(X) \subseteq \overline{\mathbb{D}}$. Suppose that*
$$\|S_v \varphi_{\omega,a}(X) S_v^{-1}\| \le K \tag{10}$$
for every unit vector v, every $|\omega| = 1$, and every $0 \le a < 1$. Then
$$\|X\| \le \max\left\{1, \frac{K}{\kappa}\right\}. \tag{11}$$

Proof. Put $t = \|X\|$. The result is immediate if $t \le 1$, so assume $t > 1$. Choose unit vectors x, y such that
$$Xx = ty.$$
Let $c_0 = \langle x, y\rangle$, put $\omega = c_0/|c_0|$ when $c_0 \ne 0$, and put $\omega = 1$ otherwise. Set
$$\widetilde{y} = \omega y.$$
Then $|\omega| = 1$, $\omega X x = t\widetilde{y}$, and $c := \langle x, \widetilde{y}\rangle = |c_0| \in [0, 1]$. In fact $c < 1$, because $c = 1$ would give $\widetilde{y} = x$ and $Xx = t\overline{\omega}x$, producing an eigenvalue of X of modulus $t > 1$, whereas $\sigma(X) \subseteq \overline{\mathbb{D}}$.

If $c = 0$, put $a = 0$. If $c > 0$, consider
$$F(a) = tc(1 + a^2) - a(1 + t^2).$$
Since
$$F(0) = tc > 0, \qquad F(1) = 2tc - (1 + t^2) \le -(t - 1)^2 < 0,$$
there is an $a \in (0, 1)$ such that
$$tc(1 + a^2) = a(1 + t^2). \tag{12}$$
Define
$$u = (I - a\omega X)x = x - at\,\widetilde{y}, \qquad w = (\omega X - aI)x = t\,\widetilde{y} - ax.$$
The matrix $I - a\omega X$ is invertible because $\sigma(X) \subseteq \overline{\mathbb{D}}$ and $a < 1$, so $u \ne 0$ and
$$\varphi_{\omega,a}(X)u = w.$$

Also $w \neq 0$, since $w = 0$ would imply $t = \|Xx\|_2 = a < 1$. Equation (12) gives

$$\langle u, w\rangle = tc(1+a^2) - a(1+t^2) = 0.$$

It also yields

$$\begin{aligned} \|w\|_2^2 - t^2\|u\|_2^2 &= a(t^2-1)\big(2tc - a(1+t^2)\big) \\ &= atc(t^2-1)(1-a^2) \geq 0. \end{aligned} \tag{13}$$

Consequently,

$$\frac{\|w\|_2}{\|u\|_2} \geq t.$$

Put $v = w/\|w\|_2$. Since $u \perp w$, the stretch S_v satisfies $S_v^{-1}u = u$ and $S_v w = \kappa w$. Testing $S_v \varphi_{\omega,a}(X) S_v^{-1}$ on u gives

$$K \geq \|S_v \varphi_{\omega,a}(X) S_v^{-1}\| \geq \kappa \frac{\|w\|_2}{\|u\|_2} \geq \kappa t.$$

Thus every case with $t > 1$ has $t \leq K/\kappa$, proving (11). $\square$

4. Transfer to scaled q-numerical ranges

We now combine the geometry and extraction into a general transfer principle.

Theorem 4.1 (Spectral-constant transfer). *Let $K \geq 1$ have the following property: for every square matrix B and every rational function h pole-free on $W(B)$,*

$$\|h(B)\| \leq K \max_{z \in W(B)} |h(z)|. \tag{14}$$

Let $n \geq 2$, $A \in \mathbb{M}_n(\mathbb{C})$, $0 < |q| \leq 1$, and let f be a rational function pole-free on $\Omega_q(A)$. With

$$r = |q|, \qquad \kappa = \frac{1+\sqrt{1-r^2}}{r},$$

one has

$$\|f(A)\| \leq \max\left\{1, \frac{K}{\kappa}\right\} \max_{z \in \Omega_q(A)} |f(z)|. \tag{15}$$

Proof. By (4), it is enough to use $\Omega_r(A)$. Put

$$M = \max_{z \in \Omega_r(A)} |f(z)|, \qquad M_\varepsilon = M + \varepsilon, \qquad X_\varepsilon = \frac{f(A)}{M_\varepsilon},$$

where $\varepsilon > 0$. The maximum exists because $\Omega_r(A)$ is compact and f is continuous there. Since

$$\sigma(A) \subseteq W(A) \subseteq \Omega_r(A),$$

the rational spectral mapping theorem gives $\sigma(X_\varepsilon) \subset \mathbb{D}$.

Fix a unit vector v and put $B = S_v A S_v^{-1}$. Lemma 2.2 gives

$$W(B) \subseteq \Omega_r(A).$$

For $|\omega| = 1$ and $0 \leq a < 1$, define the rational function

$$h(z) = \varphi_{\omega,a}\left(\frac{f(z)}{M_\varepsilon}\right).$$

The function f is pole-free on $W(B)$, and the strict inequality $M/M_\varepsilon < 1$ keeps its normalized values in $\mathbb{D}$. For every $z \in W(B)$,

$$\left|1 - \frac{a\omega f(z)}{M_\varepsilon}\right| \geq 1 - \frac{aM}{M_\varepsilon} > 0.$$

Thus h is pole-free on $W(B)$ and $|h| \leq 1$ there. Applying (14) and using similarity covariance and composition covariance of the rational functional calculus, we obtain

$$\|S_v \varphi_{\omega,a}(X_\varepsilon) S_v^{-1}\| = \left\|\varphi_{\omega,a}\left(\frac{f(B)}{M_\varepsilon}\right)\right\| \leq K.$$

Lemma 3.1 therefore gives

$$\frac{\|f(A)\|}{M_\varepsilon} \leq \max\left\{1, \frac{K}{\kappa}\right\}.$$

Letting $\varepsilon \downarrow 0$ proves (15). □

The sharp numerical-range theorem gives (14) with $K = 2$ for every rational h pole-free on $W(B)$ [2].

Proof of Theorem 1.1. Apply Theorem 4.1 with $K = 2$ and use (6). This proves (3). □

We next establish optimality of both branches of C_q.

Proposition 4.2 (Sharpness). *For every fixed $q \in \mathbb{C}$ with $0 < |q| \leq 1$, put $r = |q|$. The constant*

$$C_q = \max\left\{1, \frac{2}{\kappa}\right\}, \qquad \kappa = \frac{1 + \sqrt{1 - r^2}}{r},$$

is the least constant C such that

$$\|f(A)\| \leq C \max_{z \in \Omega_q(A)} |f(z)|$$

for every $A \in \mathbb{M}_2(\mathbb{C})$ and every rational function f pole-free on $\Omega_q(A)$.

Proof. Theorem 1.1 shows that C_q is admissible. By (4), $\Omega_q(A) = \Omega_r(A)$. The constant function $f \equiv 1$ gives the lower bound $C \geq 1$.

For the other branch, let

$$J = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

For a unit vector $x = (\xi, \eta)$, one has

$$\langle Jx, x \rangle = \bar{\xi}\eta, \qquad \rho_J(x) = |\eta|^2.$$

Put $\mu = |\xi|$ and $\nu = |\eta|$, so that $\mu^2 + \nu^2 = 1$. The identities in (6) give

$$\frac{\sqrt{1 - r^2}}{r} = \frac{\kappa^2 - 1}{2\kappa}.$$

Therefore every point z in the disk indexed by x in (7) satisfies

$$|z| \leq \mu\nu + \frac{\kappa^2 - 1}{2\kappa}\nu^2 \leq \frac{\kappa}{2},$$

because

$$\frac{\kappa}{2} - \left(\mu\nu + \frac{\kappa^2 - 1}{2\kappa}\nu^2\right) = \frac{(\kappa\mu - \nu)^2}{2\kappa} \geq 0. \tag{16}$$

Equality is attained by taking the unit vector $x = (\mu, \nu)$ with positive real coordinates

$$\mu = \frac{1}{\sqrt{\kappa^2+1}}, \qquad \nu = \frac{\kappa}{\sqrt{\kappa^2+1}},$$

For this vector, the positive boundary point is $z = \kappa/2$. Hence

$$\max_{z\in\Omega_r(J)} |z| = \frac{\kappa}{2}.$$

Since $\|J\| = 1$, taking the polynomial $f(z) = z$ yields

$$\frac{\|f(J)\|}{\max_{z\in\Omega_r(J)}|f(z)|} = \frac{2}{\kappa}.$$

Thus every admissible C satisfies $C \geq 2/\kappa$. Together with $C \geq 1$, this proves the claim. $\square$

5. Consequences and discussion

The transfer theorem accesses the base numerical-range estimate solely through (14), applied to the rank-one stretches (8) selected by the extraction mechanism. The disk-automorphism family in Lemma 3.1 supplies the orthogonal input-output pair used in (13); the identity function is one member of this family.

More generally, Theorem 4.1 turns every universal numerical-range spectral constant into an explicit constant for scaled q-numerical ranges.

For the sharp value $K = 2$, the transfer constant equals 1 precisely when $\kappa \geq 2$, or equivalently when $0 < |q| \leq 4/5$. At $|q| = 1$, one has $\kappa = 1$ and the argument reduces to the sharp numerical-range estimate. The definition of the scaled range uses $0 < |q| \leq 1$. When $n \geq 2$, unit vectors satisfying $y^*x = q$ exist for every such q.

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